

06 The reaction quotient, the equilibrium constant, and rate laws for reversible reactions

Why are we learning this?

Not all reactions proceed to complete conversion. Many reactions are **reversible**, meaning products can react to reform reactants. As a result, there is often a thermodynamic limit to how much conversion can be achieved.

To design reactors for reversible reactions, we must answer three questions:

- What composition corresponds to equilibrium?
- Which direction will a reaction proceed?
- How should equilibrium be incorporated into the rate law?

By the end of today's lesson, you should understand:

- The meaning of the equilibrium constant.
- The meaning of the reaction quotient.
- How to determine the direction of reaction.
- How equilibrium appears in reversible rate laws.
- How temperature affects equilibrium.

Course Roadmap

Introduction to CRE



General Balance Equation



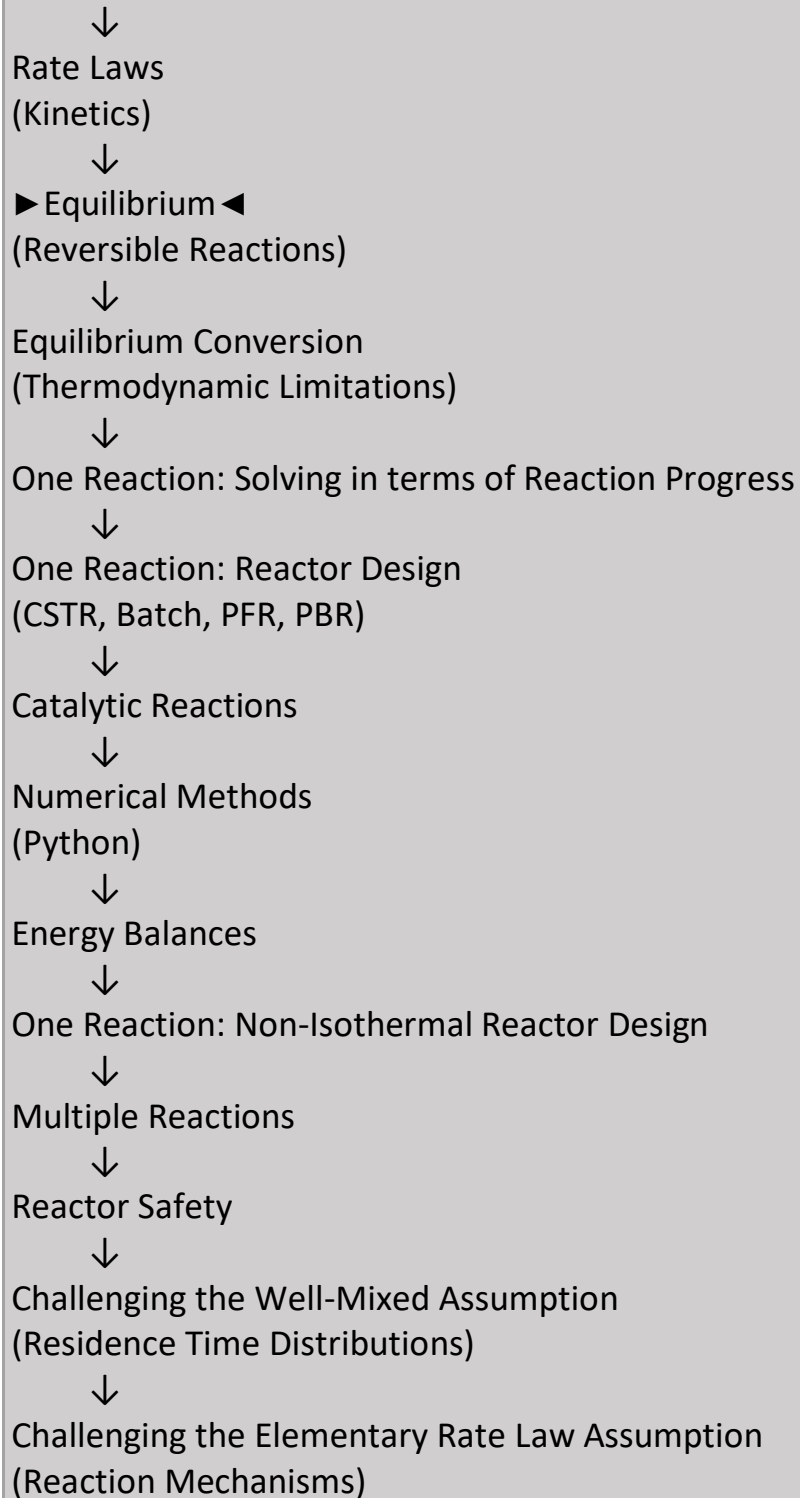
Common Reactor Models
(CSTR, Batch, PFR)



Reaction Progress
(Conversion)



Graphical Reactor Design
(Inverse Rate/Levenspiel Plots)



Big Idea

A rate law tells us: How fast a reaction proceeds.

An equilibrium constant tells us: How far a reaction can proceed.
Successful reactor design requires understanding both.

These notes are based off [this chapter](#) and [this chapter](#) from ChemLibre Texts.

Let's start with the equilibrium constant. In this class we denote
is the

equilibrium constant in terms of concentration and K_c is the
equilibrium constant

in terms of pressure.

Note that K_p is only valid for

Why Does Equilibrium Matter?

For irreversible reactions, conversion can continue increasing as long as reactants remain.

For reversible reactions, conversion has a thermodynamic limit. No matter how large the reactor becomes, equilibrium may prevent complete conversion.

This is one of the most important ideas in chemical reaction engineering.

What Does the Equilibrium Constant Mean?

The equilibrium constant tells us where equilibrium lies.

Value of K	Meaning
$K \gg 1$	Products strongly favored
$K \approx 1$	Neither side strongly favored
$K \ll 1$	Reactants strongly favored

Important

A large equilibrium constant does **not** mean the reaction is fast. It only tells us that products are strongly favored at equilibrium.

For a chemical reaction, $aA + bB \rightarrow cC + dD$, the condition for equilibrium is

$$\text{Therefore } a\mu_A + b\mu_B = c\mu_C + d\mu_D$$

Where μ are

$$\mu = \mu^0 + RT \ln a$$

Where μ^0 is

And a is

For an ideal substance, $\mu^0 =$

And $a =$

Hence, $a\mu_A + b\mu_B = c\mu_C + d\mu_D$ can be re-written

$$a\Delta G_{f,A}^0 + aRT \ln C_{A,\text{eq}} + b\Delta G_{f,B}^0 + bRT \ln C_{B,\text{eq}} = c\Delta G_{f,C}^0 + cRT \ln C_{C,\text{eq}} + d\Delta G_{f,D}^0 + dRT \ln C_{D,\text{eq}}$$

Where the subscript 'eq' indicates

This equation can be re-arranged:

$$a\Delta G_{f,A}^0 + b\Delta G_{f,B}^0 - c\Delta G_{f,C}^0 - d\Delta G_{f,D}^0 = RT \ln \frac{C_{C,\text{eq}}^c C_{D,\text{eq}}^d}{C_{A,\text{eq}}^a C_{B,\text{eq}}^b}$$

Where $a\Delta G_{f,A}^0 + b\Delta G_{f,B}^0 - c\Delta G_{f,C}^0 - d\Delta G_{f,D}^0 =$

And $\frac{C_{C,eq}^c C_{D,eq}^d}{C_{A,eq}^a C_{B,eq}^b} =$

Hence, $K_C(T) = \frac{C_{C,eq}^c C_{D,eq}^d}{C_{A,eq}^a C_{B,eq}^b} = \exp\left(-\frac{\Delta G_{rxn}}{RT}\right)$

Why Does Free Energy Appear?

Equilibrium is determined by thermodynamics. A reaction reaches equilibrium when there is no remaining thermodynamic driving force favoring either reactants or products. Free energy provides a quantitative measure of that driving force. The famous relationship

$$K_C = \exp\left(-\frac{\Delta G_{rxn}}{RT}\right)$$

connects chemical thermodynamics with equilibrium chemistry.

$\Delta G_{rxn}(T) =$

Where $\Delta H_{rxn} = d\Delta H_{f,D}^0 + c\Delta H_{f,C}^0 - a\Delta H_{f,A}^0 - b\Delta H_{f,B}^0$

And $\Delta S_{rxn} = dS_D + cS_C - aS_A - bS_B$

Values of ΔH_f^0 and S can be looked up, e.g., using [the NIST Chemistry Webbook](#).

The reaction quotient, $Q = \frac{C_C^c C_D^d}{C_A^a C_B^b}$, which is related to K_C EXCEPT THAT

The quantity $\frac{Q}{K_C}$ is known as the "reversibility" and tells us something about the

Specifically, if $\frac{Q}{K_C} > 1$,

And if $\frac{Q}{K_C} < 1$,

Q vs. K

Students often confuse these quantities.

Reaction Quotient (Q)

Uses the **current** reactor composition.

Equilibrium Constant (K)

Uses the composition that would exist **at equilibrium**.

Think of it this way:

Q tells us where we are.

K tells us where we are going.

Predicting Reaction Direction

The ratio $\frac{Q}{K}$ tells us whether a reaction wants to move toward products or reactants.

Condition	Interpretation
$Q/K < 1$	Net reaction proceeds toward products
$Q/K = 1$	System is at equilibrium
$Q/K > 1$	Net reaction proceeds toward reactants

Engineering Interpretation: This ratio tells us which direction the reactor composition wants to move.

We can use these quantities to write rate laws for reversible reactions. Consider the reaction from above: $aA + bB \rightleftharpoons cC + dD$.

The rate law is

Where $k_f =$

And $k_r =$

How are k_f and k_r related?

Consider the free energy diagram for the reaction:

Hence, $\frac{k_f}{k_r} =$

Assuming $\Delta S_{rxn,f} \approx \Delta S_{rxn,r}$, $\frac{k_f}{k_r} =$

Hence, we can re-write the rate law as $-r_A = k_f \left(C_A^a C_B^b - \frac{C_C^c C_D^d}{K_C} \right)$, which is the form that we normally use.

In terms of the reaction quotient: $-r_A = k_f C_A^a C_B^b \left(1 - \frac{Q}{K_C} \right)$. At equilibrium $Q = K_C$ hence

Why Must the Rate Law Change?

The irreversible rate laws used previously only accounted for the forward reaction. For reversible reactions:

- Reactants form products.
- Products form reactants.
- The net rate must account for both effects.

The observed rate is therefore: Forward Rate – Reverse Rate

What Happens at Equilibrium?

At equilibrium, $Q = K$ which makes $1 - \frac{Q}{K} = 0$ and therefore $-r_A = 0$.

Key Insight

At equilibrium:

- Molecules continue reacting.
- Forward reactions still occur.
- Reverse reactions still occur.
- The net reaction rate is zero.

Temperature dependence of the equilibrium constant:

$$\begin{aligned}K_C(T) &= \exp\left(-\frac{\Delta G_{rxn}}{RT}\right) = \exp\left(-\frac{\Delta H_{rxn} - T\Delta S_{rxn}}{RT}\right) \\&= \exp\left(-\frac{\Delta H_{rxn}}{RT}\right) \exp\left(\frac{\Delta S_{rxn}}{R}\right)\end{aligned}$$

Assuming ΔH_{rxn} and ΔS_{rxn} are constant with temperature,

$$\frac{K_C(T_2)}{K_C(T_1)} =$$

$$\text{So } K_C(T_2) = K_C(T_1) \exp\left[\frac{-\Delta H_{rxn}}{R} \left(\frac{1}{T_2} - \frac{1}{T_1}\right)\right]$$

Why Does Temperature Matter?

Changing temperature changes:

- Free energy,
- Equilibrium constant,
- Equilibrium composition,
- Equilibrium conversion.

Temperature is therefore one of the most important reactor operating variables.

Endothermic vs Exothermic Reactions

A useful rule of thumb:

Endothermic Reaction

$$\Delta H_{rxn} > 0$$

Increasing temperature tends to increase K.

Exothermic Reaction

$$\Delta H_{rxn} < 0$$

Increasing temperature tends to decrease K.

This idea becomes extremely important when studying non-isothermal reactor design.

Summary and Key Takeaways

Today we introduced the thermodynamic concepts required to analyze reversible reactions.

Key Ideas

- ✓ Equilibrium places a limit on achievable conversion.
- ✓ The equilibrium constant tells us where equilibrium lies.
 - Large K favors products.
 - Small K favors reactants.
- ✓ The reaction quotient tells us the current state of the reactor. Comparing Q and K predicts reaction direction.
- ✓ Reversible rate laws contain both forward and reverse contributions.
- ✓ At equilibrium, the net reaction rate equals zero, even though molecular reactions continue to occur.

✓ Equilibrium constants depend strongly on temperature.

Most Important Idea

Rate laws tell us: How fast a reaction proceeds.

Equilibrium tells us: How far a reaction can proceed.

Reactor design requires understanding both.